

Marvell Technology - Infrastructure Repurposing Policy (v2.0)

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1. Objective & Scope

This directive governs the extraction of underutilized bare-metal and containerized nodes from the legacy MERN stack environment to be reformatted as Kubernetes worker nodes for the Marvell Vision AI cluster.

2. Eligibility Gates

A node is eligible for ML Repurposing analysis if and only if its "Avg_CPU_Util" metric (as recorded in the Q2 telemetry data) is less than or equal to 0.20 (20%).

3. Compliance and Auditing

All node reallocations must be financially modeled using internal chargeback rates. Security profiles must be assessed prior to the application of the workload transitions.

4. Suitability Scoring Algorithm

For every eligible node that passes the CPU gate in Section 2, the Systems Architect must calculate the Migration Suitability Score using the following strict formula:

$$\text{Suitability_Score} = (\text{RAM_GB} * 1.5) - (\text{Avg_CPU_Util} * 10.0) - (\text{Late_Incidents} * 5.0) - (\text{Security_Penalty})$$

4.1 Variable Definitions

* RAM_GB: The total allocated RAM for the node in gigabytes.

* Avg_CPU_Util: The raw decimal value of the average CPU utilization.

* Late_Incidents: When cross-referencing the incident log, you must strictly count ONLY the incidents that occurred AFTER February 15, 2026. Incidents occurring on or before February 15 are classified as fully mitigated and evaluate to 0 in this variable.

* Security_Penalty: If the node exhibits a vulnerable payload signature in the Kali raw scan, apply a flat penalty of 20.0. If the node is clean, the penalty is 0.0.

5. Migration Execution

Nodes achieving a Suitability Score > 0.0 will be formatted and joined to the k8s-ml-worker-pool.